

# Saurabh Kishore

Hardware and EMC Engineer with 3 years of experience in DC-DC Converters, On board charger design and development within the automotive sector. Versatile in troubleshooting and optimizing performance, ensuring seamless integration within complex systems.

Bengaluru

India

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## EXPERIENCE

### **Bosch Global Software Technologies Pvt. Ltd, Bengaluru — Hardware Engineer**

August 2022 - Present

5KW DC DC converter(800-24V): Module Owner

- Worked on design of CAN FD Physical layer according to ISO-11898-2
- Resolved vehicle level CAN communication issues with respect to long cable length and higher data rates.
- Supported microcontroller port assignment of microcontroller used for DC DC converter
- Worked closely with cross domain such as verification, validation, software and system engineering to implement new requirements, modifying the existing implementation and incorporating the latest test solution in the production program to improve test, time and quality.
- Hands on experience on design and testing of 5KW DC-DC Converter PSFB topology.
- Responsible for delivery of VERDI test and report for CAN, Microcontroller and other modules.

CharCon(On board Charger + DC DC Converter ) 11KW : Module Owner

- Worked on EMI/EMC Filter design for 3KW DC DC converter.
- Component selection : worked on Filter Magnetics, collaborated with suppliers to customize Common mode and Differential mode choke (CMC, DMC).
- Tested the characteristics of CMCs and DMCs with the help of Impedance analyzer.
- Developed simulation model in LTspice to get damping of the HV and LV Filter for Active clamp flyback.

CharCon(On board Charger + DC DC Converter ) 22KW : EMI/EMC Integrator

- Hand on experience of design, testing and good understanding of Power Factor correction (PFC) boost converter and Active clamp flyback.
- Worked on Filter PCB integration for 22kW CharCon in Stuttgart, Germany.
- Performed placement study for Filters on Filter PCB from scratch

## SKILLS

Power Electronics

EMI/EMC testing

Filter Design

Simulation tools: LTspice, PLECS, MATLAB & SIMULINK

PCB Layout design

System integration

Component selection and qualification

Worst case analysis

Monte-Carlo analysis

Testing equipments:  
Oscilloscope, Network Analyzer, Power Choke tester-DPG10

## Certification

ASPICE Overview (Web-based training)

## LANGUAGES

English

Hindi

and defined PCB layout hints, EMC zone concept.

- Developed simulation models in PLECS to approximate the RAW emissions coming from Active clamp flyback and PFC.
- Resolved thermal issues of PCB Traces, CMCs and DMCs.
- Performed and resolved vehicle level failure in conducted emission (voltage method) test according to UN ECE R10, AIS-004 and CISPR-25 at GARC, Chennai
- Tested transients on LV lines according to ISO-7637-2, on HV lines according to ISO-7637-4.
- Designed Overvoltage protection module for Surge and burst for on board Charger. Good understanding of MOVs, GDT and SIDACTORS for protection.
- Worked on Fuse design for on board chargers to pass short-circuit tests.
- Good knowledge of Safety standards like ISO 17409, ISO 26262
- Resolved blockers for homologation relevant tests like Inrush current, Touch currents for On board chargers.

**Dayananda Sagar College of Engineering, Bengaluru — B.E,  
*Electrical and Electronics Engineering***

AUG 2018 - JULY 2022

Graduated with 8.8CGPA

Final year project: Battery Monitoring system for Electrical vehicle.

**Pragati public school, Kota — Senior school certification (11th  
and 12th)**

MAY 2015 - JUNE 2017

Passed with 86.6%